

Pielot Production PRD Correction

Real Chat-Based AI Revenue Agent for Local Restaurants

1. Correct Product Direction

Pielot is a real production application, not a simulated demo.

It should still feel like a custom AI chat where the owner talks naturally to Pielot, but the actions behind the chat must be real:

- Real customer/order data upload
- Real data validation
- Real customer segmentation
- Real revenue opportunity detection
- Real SMS campaign generation
- Real compliance checks
- Real owner approval
- Real campaign creation
- Real SMS sending or scheduling
- Real campaign status tracking
- Real campaign page with live metrics

The chat should be the main interface, but it should operate real backend workflows.

2. Corrected Product Summary

Pielot is a real AI SMS revenue agent for local restaurants. Owners talk to Pielot through a chat-first interface and ask for ways to increase revenue, recover slow days, bring back old customers, or create campaigns.

Pielot uses connected or uploaded POS/customer data to find revenue opportunities, generate personalized SMS campaigns, check margin and compliance guardrails, and let the owner approve campaigns before they are sent.

Once approved, Pielot actually schedules or sends the campaign through an SMS provider, creates a real campaign page, and tracks delivery, clicks, opt-outs, redemptions, attributed orders, revenue, and profit impact.

3. What “Thinking” Means in the Real Product

The app should still show thinking/progress states, but they should not be fake animations disconnected from backend work.

Each visual progress state should map to a real backend job.

Example: User asks

“Find me a way to make more money this week.”

Pielot shows progress

- Reading uploaded POS/customer data
- Validating opt-in status
- Segmenting customers
- Detecting low-demand windows
- Estimating revenue opportunity
- Checking margin guardrails
- Preparing campaign recommendation

What is actually happening

- Backend fetches customer/order data
- Data validation service checks required fields
- Segmentation service assigns customers to groups
- Revenue scan service identifies slow windows
- Campaign recommendation agent generates a structured recommendation
- Compliance guardrail service filters unsafe recipients
- Frontend displays the final insight card

The UI should make the work visible, but the work should actually happen.

4. Updated Technical Stack

Frontend

- Vite
- React

- TypeScript
- Tailwind CSS
- shadcn/ui-style components
- Framer Motion for animations
- React Router
- Zustand or React Context for state
- Recharts for analytics

Intelligence Layer

- MiniMax token plan for LLM intelligence
- Agents SDK for orchestration
- Structured JSON outputs for cards/campaigns
- Tool calls for backend actions
- Guardrails for compliance and approval

OpenAI's Agents SDK is built for applications that own orchestration, tool execution, approvals, and state; that fits Pielot because the product needs agents that can plan, call tools, use guardrails, and coordinate workflows before campaigns are approved or sent.

Backend

Recommended:

- Node.js / Express or FastAPI
- PostgreSQL or Supabase Postgres
- Redis or queue system for campaign jobs
- Background workers for SMS sending and attribution
- File storage for CSV uploads
- Webhooks for SMS events

SMS Provider

Production MVP should use one real SMS provider:

- Twilio, preferred for MVP
- Telnyx, optional alternative

Data Sources

MVP:

- CSV upload
- Google Sheets import
- Manual customer import

- Demo seed data only for testing, not core product

Future:

- Toast
 - Square
 - Clover
 - DoorDash exports
 - Uber Eats exports
 - Shopify
 - Stripe
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5. Real MVP Scope

Must Be Real

The MVP must support:

1. Restaurant profile setup
2. CSV customer/order upload
3. Data validation
4. SMS opt-in filtering
5. Customer segmentation
6. AI revenue scan
7. AI-generated campaign recommendation
8. SMS copy generation
9. Margin/profit estimate
10. Compliance checklist
11. Owner approval before sending
12. Real campaign creation
13. Real SMS scheduling/sending
14. Real campaign status page
15. Real delivery/failed/opt-out tracking
16. Real zero-state before messages send
17. Real metrics update after events arrive

Can Be Simplified

The MVP can simplify:

- POS integrations can start with CSV

- Attribution can start with promo codes and links
- Profit estimates can use owner-provided margin assumptions
- Customer segmentation can use rules + AI explanation
- Campaign performance can start with delivery, clicks, opt-outs, and manually/imported attributed orders

Should Not Be Built Yet

- Full Toast/Square/Clover integrations
 - Fully autonomous sending without approval
 - Complex ML forecasting
 - Multi-location enterprise analytics
 - Full CRM replacement
 - Billing
 - Advanced A/B testing
 - Inventory-aware campaigns
 - Weather/event automation
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6. Core Real Product Flow

Flow: Chat to Real Campaign

1. Owner logs in.
2. Owner creates restaurant profile.
3. Owner uploads customer/order CSV.
4. Pielot validates data.
5. Owner confirms SMS consent source.
6. Owner opens chat.
7. Owner asks: "Find me a way to make more money this week."
8. Pielot runs a real revenue scan.
9. Pielot shows visual progress tied to backend jobs.
10. Pielot returns a real insight card.
11. Owner clicks "Create Campaign."
12. Pielot creates a real campaign draft object.
13. Owner reviews SMS, audience, offer, and projected impact.
14. Owner clicks "Approve & Schedule."
15. Pielot schedules SMS through provider.
16. Campaign page is created.
17. Initial metrics show zero until send begins.
18. SMS provider webhooks update delivery, failure, and opt-out data.

19. Campaign page updates live.
 20. Owner sees actual campaign results.
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7. Main Chat Interface

The chat remains the main interface.

Main Screen Layout

Left Sidebar

- Pilot logo
- New chat
- Campaigns
- Customers
- Segments
- Compliance
- Settings
- Restaurant status
- Data source status
- SMS provider status

Main Chat Area

Header:

Pilot

Subtitle:

AI revenue agent for your restaurant

Status pills:

- POS data connected
- SMS provider connected
- Consent verified
- Campaign approval required

Suggested Prompts

- Find me a revenue opportunity

- Create a campaign for lapsed customers
- What should I send this week?
- Why are Tuesdays slow?
- Make this offer less discount-heavy
- Show me campaign performance

Chat Input

Placeholder:

Ask Pielot to find revenue, create campaigns, or explain your customers...

8. Real Thinking/Progress System

Every thinking card should be backed by real job status.

Component: AgentProgressCard

Required fields

- workflow_id
- workflow_type
- current_step
- steps
- status
- progress_percent
- started_at
- completed_at
- error_message
- output_card_id

Step states

- pending
- running
- complete
- failed
- needs_approval

Example: Revenue Scan Progress

Title:

Pielot is scanning for revenue opportunities

Steps:

1. Loading customer/order data
2. Checking opt-in eligibility
3. Segmenting customers
4. Finding slow revenue windows
5. Matching offers to segments
6. Estimating revenue and margin
7. Running compliance guardrails
8. Preparing recommendation

Each step should update based on backend completion, not only frontend timers.

9. Backend Workflows

Workflow 1: Data Validation

Triggered when CSV is uploaded.

Inputs

- restaurant_id
- uploaded_file_id

Actions

- Parse file
- Validate required columns
- Normalize phone numbers
- Check SMS opt-in status
- Remove duplicates
- Detect missing values
- Store customers
- Store orders
- Generate data quality score

Output

- customers_found
 - opted_in_customers
 - opted_out_customers
 - invalid_phone_numbers
 - missing_consent_count
 - data_quality_score
 - validation_warnings
-

Workflow 2: Revenue Scan

Triggered from chat or dashboard.

Inputs

- restaurant_id
- date_range
- optional user goal

Actions

- Load recent orders
- Calculate weekday/hour revenue patterns
- Detect weak revenue windows
- Segment customers
- Identify campaign opportunities
- Rank opportunities
- Estimate revenue/profit potential
- Return top recommendation

Output

- opportunity_id
 - slow_window
 - segment
 - segment_size
 - recommended_offer
 - estimated_revenue
 - estimated_profit
 - confidence
 - reasoning
-

Workflow 3: Campaign Draft Creation

Triggered when user clicks “Create Campaign.”

Inputs

- opportunity_id
- restaurant_id
- selected_segment
- recommended_offer

Actions

- Create campaign draft
- Generate SMS copy
- Generate promo code
- Estimate cost
- Estimate profit
- Run compliance checks
- Save campaign draft

Output

- campaign_id
 - status: draft
 - audience_count
 - sms_body
 - offer
 - send_time
 - promo_code
 - projected_metrics
 - compliance_status
-

Workflow 4: Campaign Approval

Triggered when user clicks “Approve & Send” or “Schedule.”

Inputs

- campaign_id
- owner_id

- approval_confirmation

Actions

- Verify owner permissions
- Re-run compliance checks
- Lock audience
- Schedule SMS job
- Create campaign recipients
- Mark campaign as scheduled

Output

- campaign_id
 - status: scheduled
 - scheduled_at
 - recipient_count
-

Workflow 5: SMS Sending

Triggered by scheduled campaign worker.

Inputs

- campaign_id

Actions

- Load campaign recipients
- Send SMS through provider
- Store provider message IDs
- Update recipient delivery status
- Mark campaign as sending/sent

Output

- sent_count
 - failed_count
 - provider_batch_id
-

Workflow 6: SMS Webhooks

Triggered by SMS provider events.

Inputs

- provider message event

Actions

- Match provider message ID to campaign recipient
- Update delivery status
- Track failed messages
- Process STOP/opt-out replies
- Update opt-out table
- Exclude opted-out customers from future campaigns

Output

- updated recipient status
 - campaign metrics updated
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Workflow 7: Campaign Attribution

MVP attribution can use:

- Unique promo code
- Unique tracking link
- Manual POS import after campaign
- Time-window matching

Inputs

- campaign_id
- promo redemptions
- order import
- click events

Actions

- Match orders to campaign
- Calculate attributed revenue

- Estimate profit impact
- Update campaign metrics

Output

- orders_attributed
 - revenue_generated
 - estimated_profit_impact
 - redemption_rate
-

10. Custom Chat Cards

Card 1: Revenue Insight Card

Shown after a real revenue scan.

Fields

- opportunity_id
- slow_window
- demand_gap
- baseline_revenue
- predicted_revenue
- segment_name
- segment_size
- recommended_offer
- estimated_revenue
- estimated_profit
- confidence
- reason

Buttons

- Create Campaign
 - Explain Insight
 - View Details
 - Ignore
-

Card 2: Campaign Draft Card

Shown after a real campaign draft is created.

Fields

- campaign_id
- campaign_name
- status: Draft
- target_audience
- audience_size
- offer
- promo_code
- send_time
- sms_preview
- projected_redemptions
- projected_orders
- projected_revenue
- projected_profit
- confidence
- compliance_checklist

Buttons

- Approve & Send
 - Schedule
 - Edit SMS
 - Change Offer
 - Change Audience
-

Card 3: Offer Comparison Card

Shown when user asks “Why this offer?”

Fields

- recommended_offer
- rejected_offers
- margin_risk
- projected_conversion
- reason

Buttons

- Keep Recommended Offer
 - Try Another Offer
 - Rewrite SMS
-

Card 4: Live Campaign Card

Shown after campaign approval/scheduling.

Fields

- campaign_id
- campaign_name
- status
- audience_size
- send_time
- promo_code
- delivery_status
- texts_sent
- delivered
- failed
- redemptions
- orders_attributed
- revenue_generated
- profit_impact
- unsubscribes

Buttons

- Open Campaign Page
 - Pause Campaign
 - Duplicate
 - View Setup
-

11. Campaign Page

URL

</campaigns/:campaignId>

Purpose

Show the real campaign object and performance.

Sections

Header

- Campaign name
- Status: Draft / Scheduled / Sending / Sent / Completed / Paused
- Campaign ID
- Created by
- Created date
- Approved by
- Scheduled send time

Campaign Setup

- Audience
- Audience size
- Segment logic
- Offer
- Promo code
- SMS body
- Send time
- Valid window
- Control/test group, optional

Live Performance

Before sending, values show zero.

- Texts sent: 0
- Delivered: 0
- Failed: 0
- Clicks: 0
- Replies: 0
- Redemptions: 0
- Orders attributed: 0
- Revenue generated: \$0
- Estimated profit impact: \$0
- Unsubscribes: 0

After webhooks/imports arrive, metrics update.

Projection vs Actual

Projection:

- Expected redemptions
- Expected orders
- Expected revenue
- Expected profit

Actual:

- Actual redemptions
- Actual orders
- Actual revenue
- Actual profit

Timeline

- Draft created
- Compliance passed
- Owner approved
- Scheduled
- Sending started
- Sending completed
- First delivery received
- First redemption received
- Results updated

Ask Pielot About This Campaign

Mini chat scoped to the campaign.

Example prompts:

- Why did we choose this audience?
- What happens if redemptions are low?
- Can we make this campaign safer?
- What should we test next?

12. Data Model

restaurants

- id
- owner_id
- name
- cuisine_type
- location
- timezone
- average_order_value
- brand_tone
- order_link
- sms_sender_name
- margin_assumption
- created_at
- updated_at

customers

- id
- restaurant_id
- first_name
- phone_number
- phone_hash
- sms_opt_in_status
- opt_out_status
- last_order_date
- order_count
- total_spend
- average_order_value
- favorite_items
- preferred_order_day
- preferred_order_time
- coupon_usage_score
- created_at
- updated_at

orders

- id
- restaurant_id
- customer_id
- order_date
- order_total

- discount_amount
- items
- channel
- promo_code
- created_at

segments

- id
- restaurant_id
- name
- description
- criteria_json
- customer_count
- created_at

campaigns

- id
- restaurant_id
- segment_id
- name
- status
- offer
- promo_code
- sms_body
- scheduled_at
- approved_by
- approved_at
- projected_revenue
- projected_profit
- actual_revenue
- actual_profit
- created_at
- updated_at

campaign_recipients

- id
- campaign_id
- customer_id
- phone_number

- provider_message_id
- delivery_status
- clicked
- replied
- ordered
- revenue_attributed
- opted_out_after
- sent_at
- delivered_at

opt_outs

- id
- restaurant_id
- phone_number
- keyword
- campaign_id
- opted_out_at

agent_runs

- id
- restaurant_id
- user_id
- workflow_type
- status
- input_json
- output_json
- started_at
- completed_at

agent_steps

- id
 - agent_run_id
 - step_name
 - status
 - started_at
 - completed_at
 - output_summary
-

13. Compliance Requirements

Pielot must be compliance-first.

Required SMS Guardrails

- Only send to opted-in customers
- Never send to opted-out customers
- Include opt-out language
- Process STOP replies automatically
- Enforce quiet hours
- Enforce frequency caps
- Require human approval before sending
- Keep approval logs
- Keep campaign audit logs
- Prevent sensitive targeting
- Prevent deceptive urgency
- Let owner pause campaigns

Default Frequency Caps

- Max 2 marketing texts per customer per week
 - Max 6 marketing texts per customer per month
 - No marketing texts before 9 AM local time
 - No marketing texts after 8 PM local time
 - No repeated campaign to same segment within 72 hours
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14. Intelligence Requirements

The MiniMax/agent layer must produce structured outputs, not just prose.

Required structured output for campaign recommendation

```
{  
  "campaign_name": "Tuesday Comeback Offer",  
  "segment_name": "At-risk weekday customers",  
  "customer_count": 1248,  
  "segment_reason": "Customers who previously ordered on weekdays but have not returned in  
21–45 days.",
```

```
"offer": "$5 off orders above $30",
"sms_body": "Pleasure Pizza: We miss you. Get $5 off your order of $30+ today from 2–6 PM.
Order here: [link]. Reply STOP to opt out.",
"send_time": "Tuesday 1:15 PM",
"projected_orders": 60,
"projected_revenue": 1680,
"projected_profit": 730,
"risk_level": "medium-low",
"compliance_status": "ready_for_approval",
"required_owner_action": "approve_before_send"
}
```

AI must not

- Hallucinate customer data
 - Send without approval
 - Target opted-out customers
 - Generate messages without opt-out language
 - Recommend offers that violate margin rules
 - Use sensitive attributes
 - Claim actual performance before data exists
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15. Real Visual Progress Requirements

Visual progress should still exist, but the frontend must subscribe to real workflow state.

Implementation

When a workflow starts:

1. Backend creates `agent_run`.
2. Backend creates `agent_steps`.
3. Frontend subscribes using polling, SSE, or websockets.
4. Each step updates as backend work completes.
5. When final output is ready, frontend renders the corresponding card.

Example

User asks:

“Find me a way to make more money this week.”

Backend creates:

- agent_run: revenue_scan
- step 1: load_data
- step 2: validate_consent
- step 3: segment_customers
- step 4: detect_opportunity
- step 5: recommend_offer
- step 6: compliance_check
- step 7: generate_card

Frontend shows these as live thinking steps.

16. Acceptance Criteria

Pilot production MVP is complete when:

1. A restaurant can create a profile.
 2. A restaurant can upload customer/order CSV data.
 3. The system validates customer data and consent status.
 4. The chat interface can trigger a real revenue scan.
 5. The system can generate at least one real revenue opportunity from uploaded data.
 6. The system can create a real campaign draft.
 7. The system can generate SMS copy with opt-out language.
 8. The system can run compliance checks.
 9. The owner must approve before sending.
 10. The system can schedule or send SMS through a real provider.
 11. The system can process delivery and opt-out webhooks.
 12. The system can create a campaign page.
 13. The campaign page shows zero before sending.
 14. The campaign page updates after real events arrive.
 15. The agent activity/progress UI reflects backend workflow state.
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17. Corrected Build Priority

Phase 1: Real MVP

Build:

- Vite React app
- Auth
- Restaurant profile
- CSV upload
- Data validation
- Customer/order database
- Chat UI
- Agent progress UI
- Revenue scan workflow
- Campaign draft workflow
- Approval workflow
- SMS provider integration
- Campaign page
- Webhook updates
- Compliance basics

Phase 2: Better Intelligence

Add:

- Better segmentation
- Offer comparison
- Margin modeling
- Promo code attribution
- Campaign recommendations
- Campaign learning

Phase 3: Integrations

Add:

- Google Sheets
 - Toast
 - Square
 - Clover
 - DoorDash/Uber Eats exports
 - Multi-location support
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Final Correct Product Principle

Pielot should feel magical, but it should not be fake.

The chat interface should make real backend workflows feel simple:

Ask Pielot → real data scan → real insight → real campaign draft → real approval → real SMS send/schedule → real campaign page → real metrics.

The “thinking” UI is not pretending. It is a visual layer over actual agent runs and backend jobs.